

Lab Assignment 1

Name:

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Subject:

Advance Database System

Section:

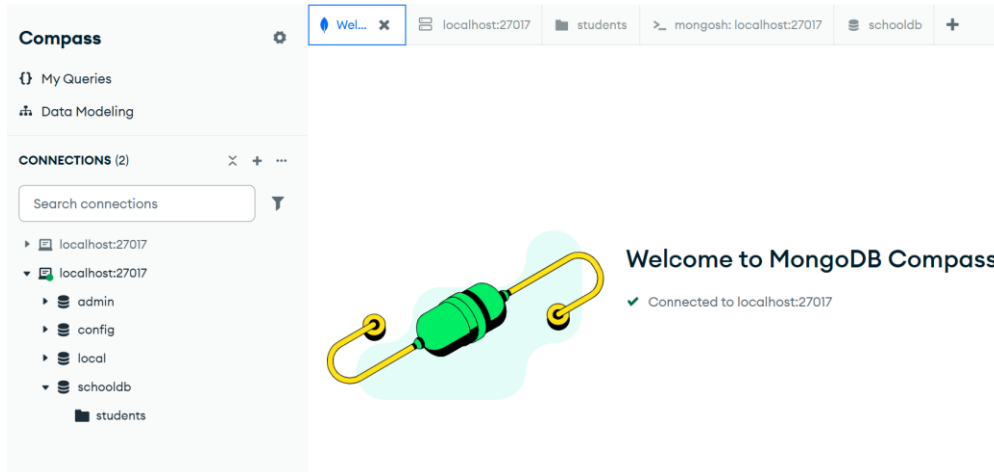
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Submitted to:

Sir Anwar Mehdi

Question 1

Downloading the MongoDB, Compass and python



```
C:\Users\SL>py -m pip install pymongo
Collecting pymongo
  Downloading pymongo-4.16.0-cp313-cp313-win_amd64.whl.metadata (10.0 kB)
Collecting dnspython<3.0.0,>=2.6.1 (from pymongo)
  Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
Downloaded pymongo-4.16.0-cp313-cp313-win_amd64.whl (959 kB)
----- 959.2/959.2 kB 5.0 MB/s 0:00:00
Downloaded dnspython-2.8.0-py3-none-any.whl (331 kB)
Installing collected packages: dnspython, pymongo
Successfully installed dnspython-2.8.0 pymongo-4.16.0

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: C:\Users\SL\AppData\Local\Programs\Python\Python313\py

C:\Users\SL>py
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb 4 2025, 15:23:48) [MSC v.1942 64 bit
Type "help", "copyright", "credits" or "license" for more information.
```

Question 2:

Commands

Database and Collection:

Database: school

Collection: students

Data:

```
{ "name": "Ali",  
  "age": 20,  
  "city": "Lahore",  
  "marks": 85 }  
  
{ "name": "Sara",  
  "age": 19,  
  "city": "Karachi",  
  "marks": 90 }  
  
{ "name": "Ahmed",  
  "age": 21,  
  "city": "Islamabad" }
```

Reading from collection

Find One Document

```
db.students.findOne()
```

```
>  
db.students.findOne()  
  
< {  
  _id: ObjectId('699b49f8da9a32a534a19086'),  
  name: 'Ali',  
  age: 20,  
  city: 'Lahore',  
  marks: 85
```

Find all documents:

```
> use schooldb
< switched to db schooldb
> db["students"].find()
< {
  _id: ObjectId('699b49f8da9a32a534a19086'),
  name: 'Ali',
  age: 20,
  city: 'Lahore',
  marks: 85
}
{
  _id: ObjectId('699b4a09da9a32a534a19088'),
  name: 'Sara',
  age: 19,
  city: 'Karachi',
  marks: 90
}
{
```

find({}).pretty()

```
db.students.find({}).pretty()

< {
  _id: ObjectId('699b49f8da9a32a534a19086'),
  name: 'Ali',
  age: 20,
  city: 'Lahore',
  marks: 85
}
{
  _id: ObjectId('699b4a09da9a32a534a19088'),
  name: 'Sara',
  age: 19,
  city: 'Karachi',
  marks: 90
}
{
  _id: ObjectId('699b4a17da9a32a534a1908a'),
  name: 'Ahmed',
```

find().count()

```
>
db.students.find().count()

< 3
```

Creation of new rows

insertOne({})

```
>
db.students.insertOne({
  "name": "Hina",
  "age": 18,
  "city": "Multan",
  "marks": 92
})

< {
  acknowledged: true,
  insertedId: ObjectId('699b5280b2ffbbdbc9d52e1e')
```

Insert many()

```
>
db.students.insertMany([
  { "name": "Bilal", "age": 22, "city": "Peshawar", "marks": 88 },
  { "name": "Ayesha", "age": 20, "city": "Quetta", "marks": 95 }
])

< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699b5247b2ffbbdbc9d52e1a'),
    '1': ObjectId('699b5247b2ffbbdbc9d52e1b')
  }
}
```

Query with conditions

```
>
db.students.find({ marks: { $gt: 85 } }).pretty()

< {
  _id: ObjectId('699b4a09da9a32a534a19088'),
  name: 'Sara',
  age: 19,
  city: 'Karachi',
  marks: 90
}
```

Querying Missing Values

```
db.students.find({ marks: { $exists: false } }).pretty()

< {
  _id: ObjectId('699b4a17da9a32a534a1908a'),
  name: 'Ahmed',
  age: 21,
  city: 'Islamabad'
```